Enterprise Network Design, Implementation and Security

Tartar Enterprise Limited

Kipchaui Ruth | CCNA Capstone Project

Built and tested in Cisco Packet Tracer



What Tartar needs

Separated departments
Logical networks for each team

Reliable site link
Head Office and Branch connected

Built-in redundancy
Less network downtime

Room to grow
Scalable addressing for expansion

Protected access
No unauthorized devices or rogue DHCP servers

Controlled management
Remote access for authorized IT only

Project approach

Design and connect

Head Office and Branch linked by a point-to-point WAN with EIGRP routing

Segment and address

VLANs, subnetting and inter-VLAN routing, with DHCP for selected VLANs

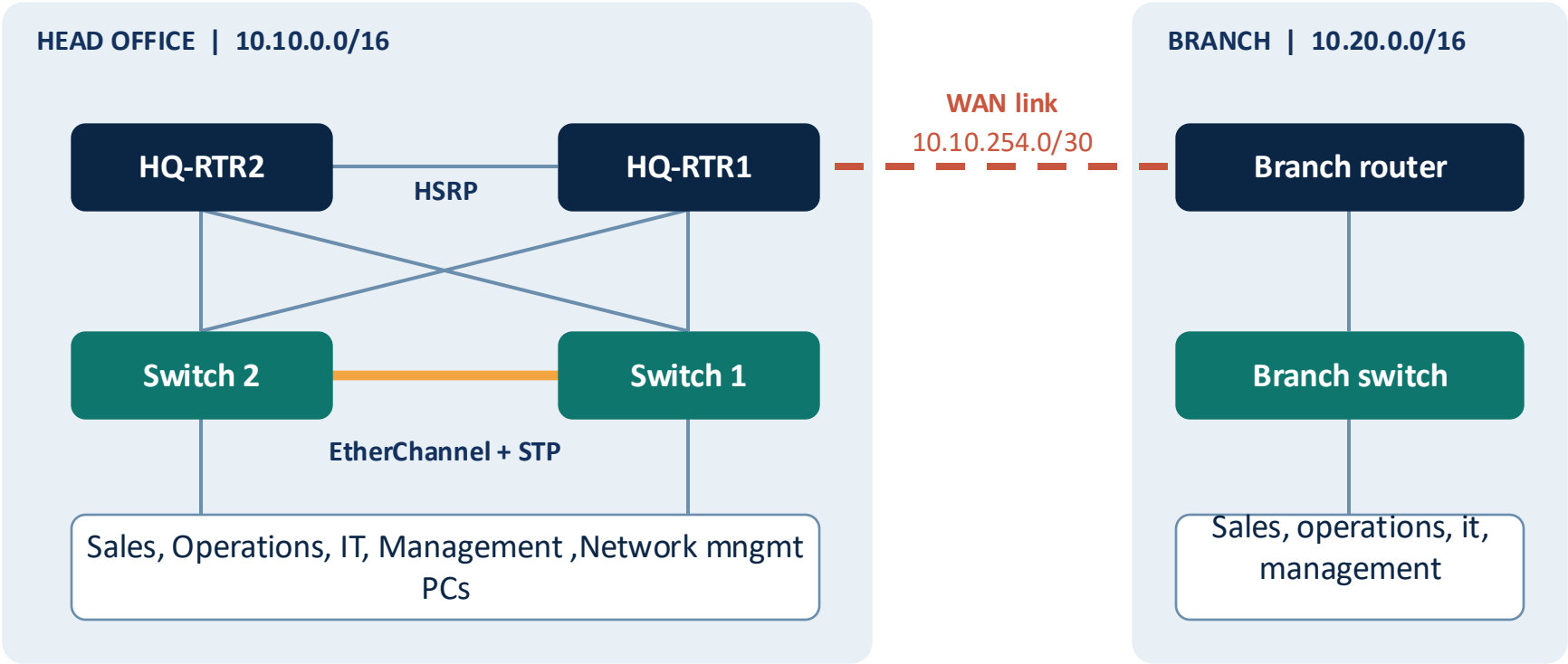
Secure

ACLs, port security, DHCP snooping, SSH and device hardening

Stay resilient

HSRP, EtherChannel and Spanning Tree for gateway and link redundancy

Network topology



Departments on separate VLANs

VLAN
10
Sales
HQ 20 hosts
Branch 15 hosts

VLAN
20
Operations
HQ 20 hosts
Branch 15 hosts

VLAN
30
IT
HQ 10 hosts
Branch 8 hosts

VLAN
40
Management
HQ 8 hosts
Branch 5 hosts

VLAN
99
Network Management
HQ 6 hosts
Branch 4 hosts

Switches get management addresses in VLAN 99 and use its gateway as their default gateway.

IP addressing plan

10.0.0.0/8
Enterprise private block

10.10.0.0/16
Head Office

10.20.0.0/16
Branch Office

10.10.254.0/30
WAN point-to-point

HEAD OFFICE

VLAN	Department	Subnet	Gateway
10	Sales	10.10.10.0/27	10.10.10.1
20	Operations	10.10.20.0/27	10.10.20.1
30	IT	10.10.30.0/28	10.10.30.1
40	Management	10.10.40.0/28	10.10.40.1
99	Network Mgmt	10.10.99.0/28	10.10.99.1

BRANCH OFFICE

VLAN	Department	Subnet	Gateway
10	Sales	10.20.10.0/27	10.20.10.1
20	Operations	10.20.20.0/27	10.20.20.1
30	IT	10.20.30.0/28	10.20.30.1
40	Management	10.20.40.0/29	10.20.40.1
99	Network Mgmt	10.20.99.0/28	10.20.99.1

Routing, WAN and DHCP

EIGRP	Routers automatically learn the networks available at both sites
Inter-VLAN routing	Departments communicate only as company policy allows
Point-to-point WAN	A 10.10.254.0/30 link joins the Head Office and Branch routers
DHCP	Gives clients an IP address, subnet mask, default gateway and DNS information on at least one department VLAN

Network security

Extended ACLs

Restrict traffic between departments, e.g. Sales to IT, management

Port security

Only approved devices can use access ports

DHCP snooping

Trusted ports only, so rogue DHCP servers are blocked

SSH

Encrypted remote management, limited to authorized IT personnel

Device hardening

Encrypted passwords and login banners on every device

Redundancy and loop prevention

HSRP

Gateway redundancy

If the primary router is unavailable, the backup provides gateway services.

.1 virtual | .2 RTR1 | .3 RTR2

EtherChannel

Link redundancy

Combines multiple physical switch links into one logical link for resilience and capacity.

Spanning Tree

Loop prevention

Stops Layer 2 switching loops across the switched network.

Testing and verification

Connectivity

- Same-VLAN pings
- Permitted VLAN pings
- Head Office to Branch
- VLANs and trunks
- DHCP allocation

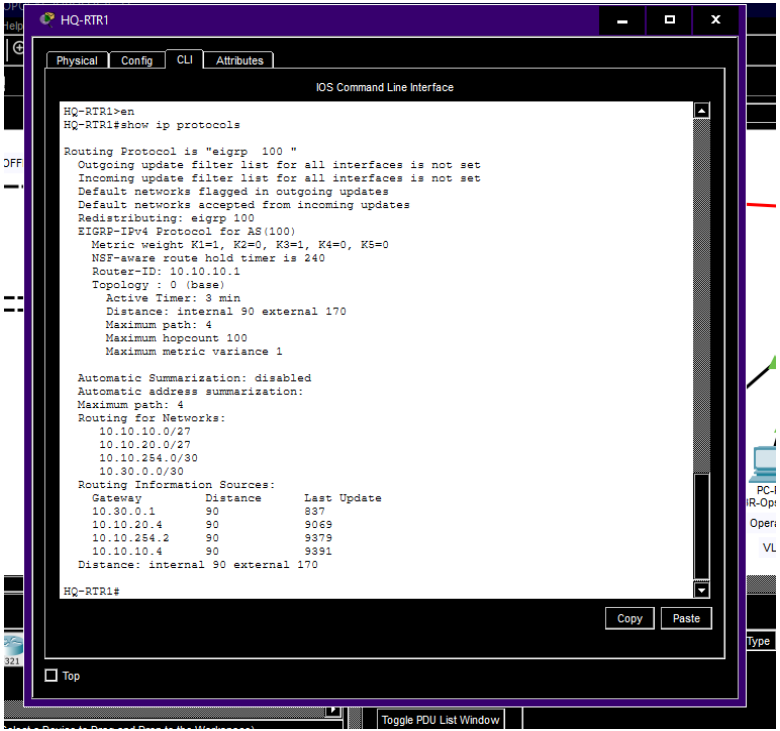
Routing and security

- EIGRP neighbors
- Routing tables
- ACL restrictions
- Port security
- DHCP snooping

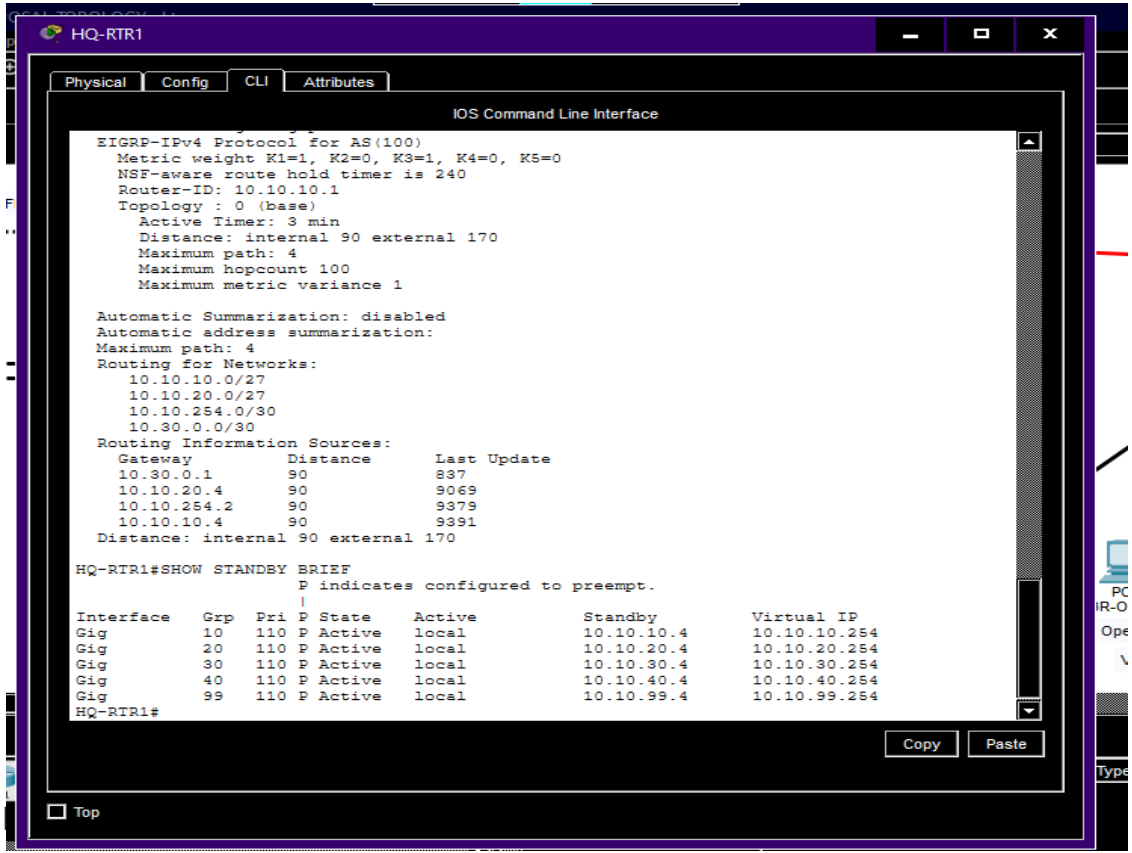
Access and resilience

- SSH access
- HSRP failover
- EtherChannel
- Spanning tree

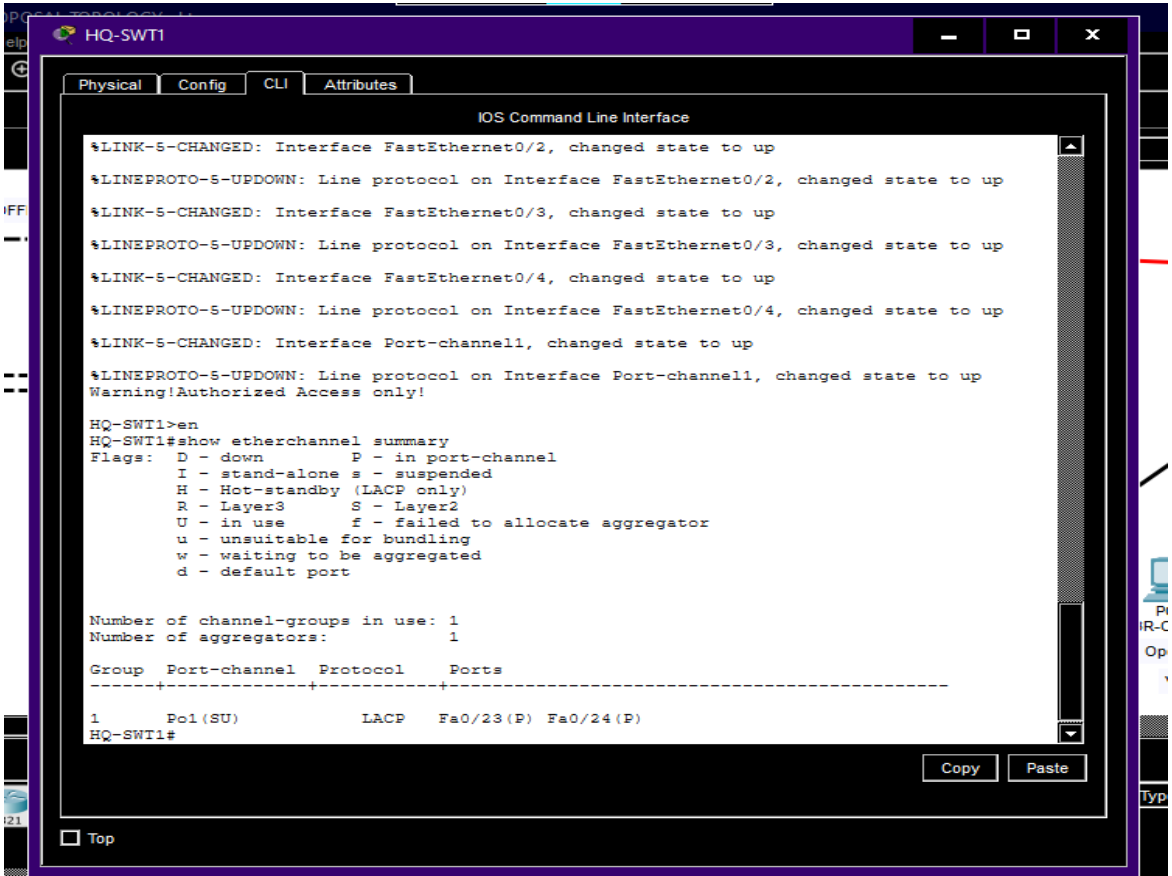
Verification screenshots
EIGRP verification



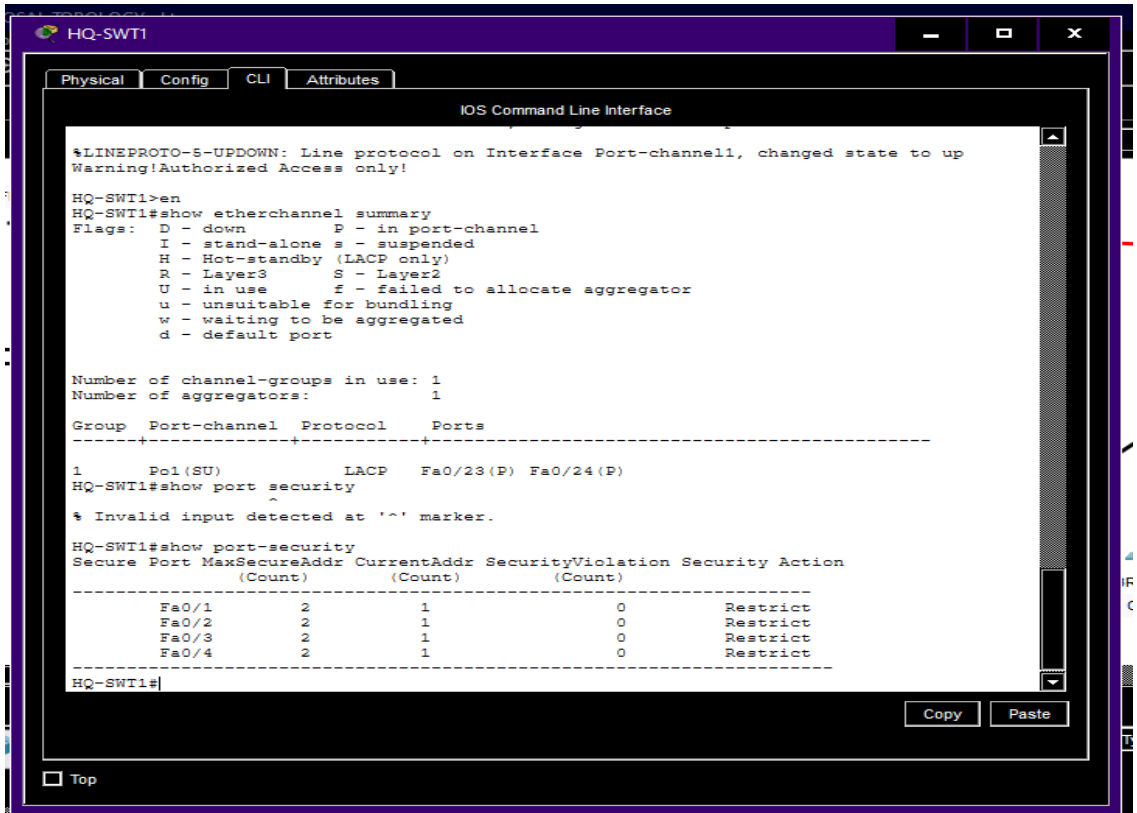
HSRP verification



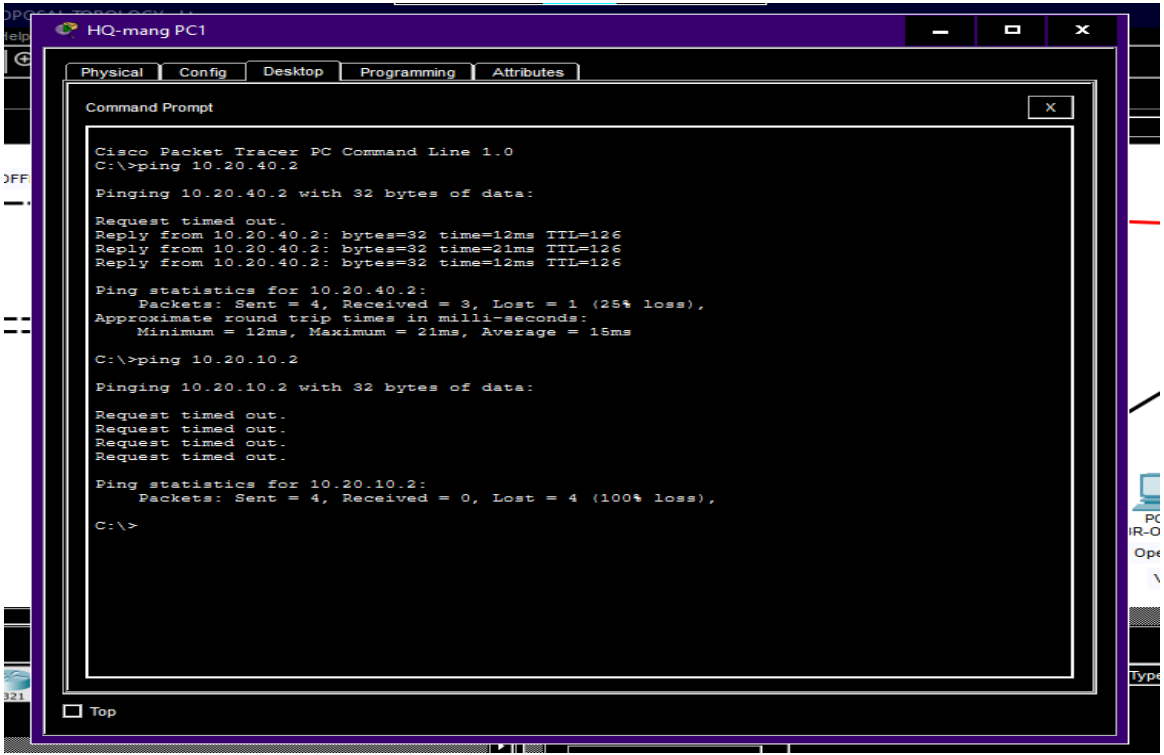
Ether channel verification



Port security verification



Ping verification



Expected outcomes

- Reliable Head Office to Branch connectivity
- Departments separated by VLANs
- Efficient IP addressing
- Dynamic routing with EIGRP
- Protection from unauthorized and rogue devices
- Secure remote management through SSH
- Gateway and link redundancy
- Loop-free, scalable foundation for growth